

Hariprakash Karthikeyan

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SUMMARY

Machine Learning Engineer / Data Scientist building scalable statistical and machine learning models for business and product problems. Experienced in predictive modeling, experimentation, time-series analysis, and end-to-end ML pipelines.

EDUCATION

University of Rochester

MS, Data Science

Rochester, New York

Aug 2024 – Dec 2025

- **Relevant Coursework:** Data Science at Scale, Statistical Machine Learning, Network Science Analytics, Database Systems

SRM Institute of Science and Technology

Bachelor of Technology, Electronics and Communication Engineering

Chennai, India

Aug 2020 – May 2024

SKILLS

Programming & Querying: Python, SQL, R

Machine Learning & Statistics: Statistical Modeling, Predictive Modeling, Supervised Learning, Time-Series Analysis, Experimental Design

ML Frameworks & Libraries: scikit-learn, TensorFlow, PyTorch

NLP & LLMs: NLP Pipelines, Embeddings, Retrieval-Augmented Generation (RAG)

Data Systems: Spark, Databricks, SQL Databases

MLOps & Deployment: ML Pipelines, Model Deployment, MLflow

Visualization & Tools: Matplotlib, Plotly, Git, Linux

Cloud & Process: AWS, Agile / Scrum

EXPERIENCE

Butler/Till

Rochester, New York

Data Scientist

Aug 2025 – Dec 2025

- Designed, trained, and evaluated hierarchical Bayesian Media Mix Models to quantify channel effectiveness and support media **optimization across ~170 DMAs**.
- Built scalable modeling pipelines incorporating adstock, saturation, and geographic heterogeneity, enabling scenario analysis and **ROI estimation across multiple budget** and channel configurations.
- Validated media mix model performance using historical holdout periods, **achieving RMSE = 0.84 relative to baseline** assumptions and ensuring stable estimates across ~170 DMAs.

Buffalo Solar

Buffalo, New York

Data Science Intern

June 2025 – July 2025

- Built predictive and statistical time-series models to compare **expected vs. actual energy production across 468 sites**, supporting anomaly detection and performance monitoring.
- Developed and deployed modeling and analytics pipelines into a production Streamlit application **used by engineering and operations teams for ongoing monitoring**.
- Iteratively refined model logic, thresholds, and **validation checks using historical** data to improve evaluation rigor, stability, and real-world usability, **achieving ~90% alignment** across 468 sites and improving real-world monitoring reliability.

University of Rochester Medical Center

Rochester, New York

Research Assistant

Feb 2025 – May 2025

- Applied statistical modeling and machine learning techniques to **large-scale EHR time series data across 2,000+** patients to answer open-ended clinical research questions.
- Built feature engineering and NLP pipelines to extract structured clinical variables from unstructured text, curating a **validated cohort of 845 patients** for downstream predictive modeling.
- **Developed time-aligned data and modeling workflows** reviewed and validated by principal investigator and clinicians.

PROJECTS

End-to-End Distributed Sentiment Analytics (Databricks, Spark)

- Designed and implemented scalable machine learning pipelines to process and model large-scale text data (100k+ records) using Spark and Delta Lake.
- Tracked experiments, metrics, and artifacts using MLflow to ensure reproducibility, performance monitoring, and deployment readiness.

Urban Mobility & Demand Analytics (NYC Taxi Data)

- Applied statistical modeling and time-series analysis to large-scale transportation data (~60GB) to model demand patterns and temporal dynamics.
- Built aggregated features and analytical summaries to support predictive analysis and operational insights.